

M03 Quick Reference - RC1

Retrieval sheet, not a substitute for practice

DE03-01 INNER JOIN and LEFT JOIN

INNER keeps matched rows. | LEFT keeps every left-side row plus matches.

DE03-02 Cardinality, row multiplication and join validation

Check key uniqueness before joining. | One-to-many joins legitimately multiply left rows.

DE03-03 UNION and UNION ALL

UNION ALL keeps every row. | UNION performs duplicate elimination.

DE03-04 String functions for cleaning and standardization

TRIM removes surrounding whitespace. | UPPER/LOWER normalize case.

DE03-05 Dates, timestamps and time analysis

DATE and timestamp types are different. | Date differences answer elapsed-time questions.

DE03-06 Subqueries and nested logic

Scalar subqueries return one value. | Correlated subqueries run in relation to outer rows.

DE03-07 CTEs and readable multi-step SQL

WITH defines named query steps. | Each CTE should have a clear grain.

DE03-08 Window functions and PARTITION BY

GROUP BY reduces rows; windows usually preserve them. | PARTITION BY creates calculation groups.

DE03-09 ROW_NUMBER, ranking and LAG/LEAD

ROW_NUMBER needs deterministic ordering. | RANK/DENSE_RANK handle ties differently.

DE03-10 Advanced aggregation and multi-grain reasoning

Name the grain after every major step. | Ratios need numerator and denominator at compatible grains.

DE03-11 PostgreSQL setup, schemas and dialect awareness

PostgreSQL uses databases and schemas as namespaces. | search_path affects unqualified object lookup.

DE03-12 Constraints and transactions

PRIMARY KEY enforces row identity. | FOREIGN KEY protects references when enabled/enforced.

DE03-13 Indexes and EXPLAIN basics

Index keys should match actual filter/join patterns. | More indexes are not automatically better.

DE03-14 Views and reusable database logic

A view stores query definition, not ordinary copied rows. | Views can centralize approved logic.

DE03-15 Integrated unseen SQL challenge

Inspect schema first. | Plan joins and grain on paper.

DE03-16 Timed SQL interview set and explanation

Time-box the first attempt. | Prefer simple correct queries over clever ones.